Modeling the Effects of Assimilable Nitrogen and Temperature on Fermentation Kinetics in Enological Conditions

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Abstract: We propose a dynamic model of alcoholic fermentation in wine-making conditions. In this model, the speed at which CO_2 is released is related to the effects of the main factors involved in fermentation in wine-making conditions: temperature (which can vary within a predefined range) and nitrogen additions (which must not exceed the maximal authorized level). The resulting model consists of ordinary differential equations including numerous parameters that need to be identified and important interactions between explicative variables. These parameters were identified by uncoupling the effects of variables during specific experiments. The results were validated on another series of experiments in different conditions. © 2004 Wiley Periodicals, Inc.

Keywords: wine; mathematical model; nitrogen; temperature; fermentation

INTRODUCTION

Wine production units are continually undergoing modernization. Improvements are particularly important for the fermentation step, which, despite being a major part of the wine production process, is still carried out empirically. If this step could be made more efficient, we could improve the management of tanks and energy. Indeed, prediction and optimization of the duration of the fermentation and minimization of the energy required to cool one tank, or the entire winery, are currently major challenges for enologists. The Identification and Automatic Control tools are valuable new instruments that may help enologists attain these objectives, by means of on-line monitoring of the fermentation. If these techniques are to be efficient, they require a good mathematical model of the fermentation process in enological conditions.

We describe here the construction of a model designed to meet the specific needs of fermentation control. Yeast metabolism is highly complex and is not yet fully understood. Thus, we did not attempt to construct a model including all the metabolic pathways of yeast. However, as we wished to ensure that the model could be used in a wide range of enological conditions, we avoided ''black box models.'' We therefore tried to develop a model of yeast activity likely to predict macroscopic phenomena useful in fermentation control. The model presented herein is based on physiological considerations and takes into account the main phenomena directly affecting kinetics in these specific conditions, including the effects of: (i) nitrogen in yeast synthesis (growth phase) and the activation of metabolism (stationary phase); (ii) the transport of sugar into yeast cells; (iii) ethanol inhibition; and (iv) temperature, including anisothermal conditions, which we know to be very common in wine-making.

Other researchers have attempted to model alcoholic fermentations (Remedios Marín, 1999). Unfortunately, most of the models proposed for the description of alcoholic fermentations are poorly adapted to enological conditions. This is particularly true as far as the relationship between two essential variables affecting fermentation kinetics is concerned. Assimilable nitrogen is generally not taken into account at all and the description of temperature effects is generally unsatisfactory. These two variables are particularly important because their manipulation enables the enologist to control the reaction. We chose to use a more general approach in the development of our model. We began by constructing a mathematical model taking into account recent advances in physiological knowledge (principally concerning the effect of nitrogen). We validated this model qualitatively and quantitatively by means of specific experiments. Toward this end, we carried out more than 100 fermentations with on-line monitoring of the rate of CO_2 production.

The model we built is an explanatory model based on physiological information and observations. An original feature of this model was the use of specific CO_2 production rates, which are directly correlated with the activity of the cell.

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We also took anisothermal conditions into account. In fact, as pointed out by Sablayrolles and Barre (1989, 1993a, 1993b), the temperature during the first part of fermentation affects the kinetics of the rest of the reaction.

We validated our model qualitatively by checking that the mathematical structure is consistent with our assumptions and with previous publications. We then carried out statistical identification of the essential parameters and validated the model quantitatively.

MATERIALS AND METHODS

Culture Conditions

Yeast Strain

The yeast strain used was *Saccharomyces cerevisiae* K1 ICV-INRA. Bioreactors were inoculated with 10^6 cells/mL active dry yeast.

Medium

We used the synthetic medium described by Bely et al. (1991). The ammonium and amino acid composition of this medium simulates the nitrogen content of a standard grape must, including proline, which is poorly assimilated. Total nitrogen concentration varied from 89 to 709 mg/L, corresponding to assimilable nitrogen concentrations of 71.4 to 571 mg/L, with the same ratio of nitrogen sources used in each case. The only sugar in the medium was glucose, which was present at concentrations of 200 or 280 g/L.

Nitrogen Additions

Next, 300 mg/L $(NH_4)_2HPO_4$ was added in some experiments. This supplement provides 63 mg/L of atomic nitrogen, entirely assimilable, corresponding to the maximum nitrogen addition permitted in wine-making. Nitrogen was added at various points in the fermentation, and with various dynamics.

Bioreactors

Small bioreactors (1.1 L) with fermentation locks were used. The fermentation mixture was stirred mechanically throughout the experiment.

Monitoring of Fermentations

CO_2

CO_2 release was determined by automatic measurement of bioreactor weight loss every 20 min. The validity of this technique as a method of monitoring fermentation has already been demonstrated (Bezenger et al., 1985; El Haloui et al., 1987). CO_2 production rate (dCO_2/dt) was calculated by polynomial smoothing of the last ten measurements of CO_2. The numerous acquisitions of CO_2 release and the precision of weighing (0.1 to 0.01 g) made it possible to calculate CO_2 production rate with a high level of precision.

Temperature Control Unit

Most fermentations were carried out in isothermal conditions, with temperatures within the enological temperature range (18° to 30°C). A thermal model was applied to some experiments in order to simulate the thermal behavior of tanks of the size used in industry (Sablayrolles et al., 1987).

Cell Population

Cells were counted using an electronic particle counter (Coulter ZBI), fitted with a probe with a 100-μm aperture. The cell size distribution was determined with a Coultronics C56 Channelyser.

Yeast Viability

Yeast viability was measured by an epifluorescence method (King et al., 1981).

Glucose

The initial (S_{init}) and final (S_{fin}) glucose concentrations were determined with dinitrosalicylic reagent (Miller, 1959). During fermentation, glucose concentration (S) was estimated from the amount of CO_2 released. The progress of fermentation was expressed as $1 - S/S_{init}$.

Assimilable Nitrogen

Ammonium concentration was determined by enzyme assay and total nitrogen concentration was determined by distillation and back-titration according to the Kjeldahl mineralization method (Bergmeyer and Beutler, 1985). Assimilable nitrogen was calculated as the difference between total nitrogen concentration in the must and at the point where the progress of fermentation reached 80% (minimum nitrogen level) (Sablayrolles et al., 1996).

Computing

We used the MATLAB and SPLUS software, with the NLS2 library for identification of parametric nonlinear regression models (Huet et al., 1996).

MODELING

What Kind of Model?

The mechanisms involved in fermentation are complicated and extend beyond the simple transformation of glucose into alcohol. The construction of an ''exact'' model of

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fermentation is therefore unrealistic. In the context of current technology, the model must meet two requirements. The first concerns the monitoring of the fermentation: it must be possible to estimate the state of the bioreactor in real time based on the model and measurable quantities. The second requirement relates to fermentation management (tank or energy management): it must be possible to predict fermentation dynamics from quantities that the enologist may change.

The main measurable quantities considered herein involve the amount of CO_2 released (or the rate of CO_2 production, dCO_2/dt). We considered the enologist to have two means of acting on the fermentation dynamics (corresponding to those that have the greatest impact on kinetics): control of the temperature in the tank and the addition of ammoniacal nitrogen during the fermentation. Addition of nitrogen is not usually taken into account in the modeling of fermentations in enological conditions. Nevertheless, this addition, which is one of the few allowed by law, is of major interest. In fact, as pointed out by Bely et al. (1990a) and Sablayrolles et al. (1996), it has a dramatic impact on fermentation kinetics by increasing the fermentation rate, lowering the fermentation duration, and (when combined with oxygenation) avoiding problems of sluggish or stuck fermentations.

The aim of this work was to generate a dynamic model of fermentation, with temperature and added ammoniacal nitrogen as the input variables and rate of CO_2 production as the output variable. The model we constructed is summarized in the following series of equations:

$$\begin{cases} \frac{dS}{dt} = -X(t)\, v_{ST}[S(t),E(t),T(t)]\, N_{ST}[N_{\max}(t) - N(t),X(t),T_{ucd}(t)] & S(0) = S_{init} \\ \frac{dN}{dt} = -X(t)\, v_N[N(t),E(t),T(t)] & N(0) = N_{init} \\ \frac{dX}{dt} = k_1[T(t)]\, X(t) \left[1 \frac{-X(t)}{X_{\max}(N_{init})}\right] & X(0) = X_{init} \end{cases} \quad (1)$$

where $t \geq 0$ and t stands for time; $X(t)$ is the cell population in the tank (L^{-1}), the maximum for this variable being denoted by $X_{\max}$ (this variable is a function of N_{init}); $S(t)$ is the glucose concentration in the tank (g/L^{-1}); $E(t)$ is the ethanol concentration in the tank (g/L^{-1}); $N(t)$ is the assimilable nitrogen concentration in the tank (g/L^{-1}) [$N_{\max}(t) = N_{init}+N_{add}(t)$, where N_{add} is the amount of nitrogen added, if any]; $T(t)$ is the temperature in the tank (°C); $T_{ucd}(t)$ is the temperature to the end of the growth phase (°C), which remains constant in the stationary phase; and $k_1(T)$ is the growth rate of the population in the exponential growth phase.

The functions v_{ST}, N_{ST}, and v_N are described in what follows. It should be noted that the concentrations of glucose and ethanol, $S(t)$ and $E(t)$, are deduced directly from the amount of CO_2 released, $CO_2(t)$. Assuming Gay–Lussac-like relationships (constant yield of CO_2 and ethanol throughout the process), we added the following equations to the previous model [Eq. (1)]:

$$\begin{cases} S(t) = S(0) - 2.17\,CO_2(t) & CO_2(0) = 0 \\ E(t) = 0.464[S(0)-S(t)] \end{cases} \quad (2)$$

where $CO_2(t)$ is the amount of CO_2 released (g/L). See El Haloui et al. (1987) for a description of the way in which the coefficients in Eq. (2) were obtained.

Comments on the Model

Before justifying the model [Eq. (1)], we briefly describe the principal ideas underlying its construction.

We investigated the effects of both initial nitrogen concentration and added nitrogen on fermentation kinetics. It is known that the rate of fermentation increases with initial nitrogen content (Bely et al., 1990). Nitrogen exerts this effect on reaction rate by affecting both population growth and the activity of the yeast (i.e., specific rate of fermentation). This makes it necessary to model both yeast growth and the average activity of a single yeast cell in its environment. The models [Eqs. (1) and (2)] satisfy these two requirements.

Growth Modeling

Many models of biological reactions are based on mass conservation. However, alcoholic fermentation in enological conditions must be dealt with differently. A model based on the conservation of mass between assimilable nitrogen and the number of yeast cells, for example, would generate incorrect results. Although nitrogen is the main limiting nutrient for growth, the ratio of initial nitrogen content to maximum population size is not a constant, but a nonlinear function depending on the amount of assimilable nitrogen (Fig. 1). Nitrogen is usually the first of all the substrates required for the growth of the yeast to be depleted, and yeast cells compete for nitrogen. The explanatory modeling of

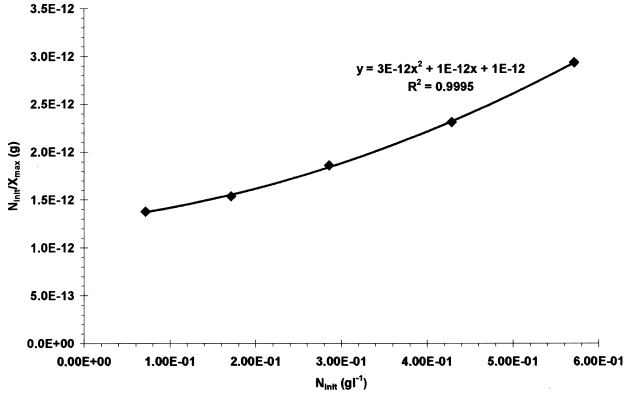


Figure 1. Changes in nitrogen concentration of each cell during the stationary phase for musts with different initial nitrogen concentrations.

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growth therefore results in a complex mathematical system derived from a partial differential equation. For the sake of simplicity, we chose not to include this complex equation in the initial complete model of fermentation (we are currently developing a model taking this relationship into account). The model of cell growth presented here is therefore not explanatory. It is instead a logistic model, of the type commonly used for population growth. Its differential form is given in the system of Eq. (1). A closed form of this differential equation exists if temperature is assumed to be constant:

$$\begin{aligned} X(t) &= X_{init}\frac{e^{k_1(T)(t-t_{latency})}}{1+\frac{X_{init}(e^{k_1(T)(t-t_{latency})}-1)}{X_{\max}(N_{init})}} && \text{if } t > t_{latency} \\ X(t) &= X_{init} && \text{else} \end{aligned} \quad (3)$$

where $t_{latency}$ is the time at which cell division begins.

Yeast Activity

Yeast activity is implicitly described by four subsystems (Fig. 2): glucose transport; glycolysis; nitrogen transport; and synthesis of glucose transporters.

Glucose Transport

v_{ST} describes the activity of a glucose transporter, by means of a facilitated diffusion mechanism, with ethanol inhibition. It is of the following form:

$$v_{ST}(S,E,T) = \frac{k_2(T)S}{S+K_S+K_{SI}SE^{\alpha_S}} \quad (4)$$

where K_S, K_{SI}, and α_S are constants, and k_2 is a function of temperature.

Glycolysis

Glycolytic enzymes have high levels of activity and are not inhibited by ethanol (Ingram and Buttke, 1984; Larue et al., 1984).

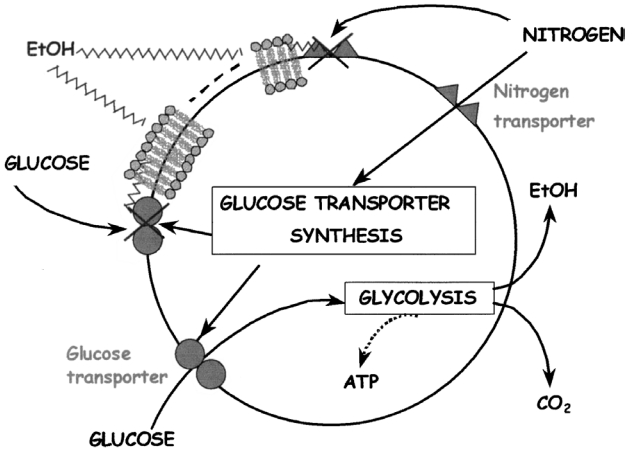


Figure 2. Simplified diagram of the activity of a yeast.

Process of Nitrogen Assimilation

v_N is related to the nitrogen absorption mechanism, which is strongly inhibited by ethanol (Leao and van Uden, 1983):

$$v_N(N,E,T) = \frac{k_3(T)N}{N+K_N+K_{NI}\,NE^{\alpha_N}} \quad (5)$$

where K_N, K_{NI}, and α_N are constants and k_3 is a function of temperature.

Synthesis of Glucose Transporters

The function $N_{ST}[N_{\max}(t) - N(t),X(t),T_{ucd}(t)]$ represents the mean number of transporters in a yeast. It models the mechanism by which the cell transforms a fraction of the absorbed nitrogen into glucose transport proteins. This relationship implies that N_{ST} depends on the nitrogen assimilated by a single cell [i.e., on the quantity $(N_{\max}(t) - N(t))/X(t)$], and environmental conditions during assimilation (principally the temperature, T_{ucd}).

QUALITATIVE VALIDATION

Before identifying the parameters of the model, it is important to validate the mathematical structure of the model [Eq. (1)] with respect to the assumptions and information used to construct it. With this in mind, we verified that no previously published work contradicted this structure and, if no published data were available, we ran specific experiments.

Glucose Transport as Limiting Step of Fermentation

We assumed that the activity of the yeast was determined principally by the activity and number of hexose transporters in the plasma membrane. Indeed, measured intracellular free glucose concentration has been shown to be very low, even if the external concentration of glucose is very high (Becker and Betz, 1972), demonstrating that glucose is instantaneously metabolized by the glycolytic enzymes of yeast. Other studies have shown that glucose transport, assayed with nonmetabolizable glucose analogs such as D-xylose or D-arabinose, displays maximum rates in the range of glycolytic flux (Serrano and De la Fuente, 1974) or of the measured fermentation rate (Salmon et al., 1993). We therefore assumed that glucose uptake was the rate-limiting step of fermentation [even if it is sometimes assumed to be tightly coupled to hexokinase (Bisson, 2000), so that glucose uptake never exceeds its phosphorylating capacity. Nevertheless, Eqs. (1)–(2) can be interpreted according to this last phenomenon, as it does not change their expressions].

The equations given correspond to an enzymatic reaction in which the permease in the plasma membrane transfers the solute across the membrane by undergoing reversible conformational changes that expose its solute-binding site alternately on either side of the membrane (André, 1995), facilitating the diffusion of the substrate in the cell.

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Inhibitory Effect of Ethanol on Glucose Transport

There are two main interpretations of the effect of ethanol on yeast activity, and we needed to decide which was the most appropriate for our model. First, ethanol inhibition may be interpreted as a ''mechanical'' effect on the capacity of the cell to take up glucose and to transport it across the membrane. The second interpretation, based on the adaptation of the yeast to its environment, is more complex. It links the decline in activity as fermentation progresses to a transporter inactivation mechanism (involving endocytosis-mediated internalization and degradation in the vacuole), as the transporters cease to be adapted to the composition of the plasma membrane (Bisson, 2000).

We decided that the hypothesis of a direct ''mechanical'' inhibitory effect of ethanol on transporters was the most appropriate, for two main reasons. First, yeast activity rapidly decreases after the addition of ethanol to the medium. Experiments performed with 2%, 4%, or 6% ethanol added at various timepoints during fermentation (Ansanay-Galeote et al., 2001) have shown that activity rapidly decreases to a level corresponding to the new amount of ethanol. Thus, fermentation rate is primarily a function of the concentration of alcohol in the medium. It should be noted that the addition of alcohol does not result in an increase in cell death. Second, many investigators have claimed that inhibition of the hexose transport system by ethanol may be attributed to changes in the lipid environment of the transporters in the plasma membrane, resulting in a conformational change unfavorable to hexose uptake, thereby modifying transporter activity (Leao and van Uden, 1982a, 1982b; Loureiro-Dias and Peinado, 1982). The plasma membrane thus undergoes dramatic changes in composition to maintain ethanol tolerance and to enable the hexose transport proteins to work efficiently in the changing microenvironment (Bisson et al., 1993). These two arguments led us to model the function, v_{ST}, with a noncompetitive effect of ethanol inhibition.

Importance of Nitrogen: Effect on Glucose Transporter Synthesis

Another assumption concerned the role of nitrogen in yeast metabolism both at the beginning of the fermentation and following its addition during fermentation. Nitrogen is involved in the kinetics of fermentation and, as nitrogen compounds are often present only in small amounts in the must (Dukes et al., 1991), they are usually the limiting factor for yeast growth and activity (Fig. 3) (Agenbach 1977; Bely et al., 1990; Bezenger et al., 1986). Moreover, the addition of ammonium ions during the stationary phase leads to partial reactivation of the hexose transport systems, increasing fermentation rate (Salmon, 1989). We therefore assumed that, during the growth phase (or during nitrogen addition), some of the assimilated nitrogen is systematically directed to a ''pool'' of nitrogen intended for use in the synthesis of sugar transporters. This mechanism is modeled by N_{ST}, the number of sugar transporters, which is a nonlinear function of the

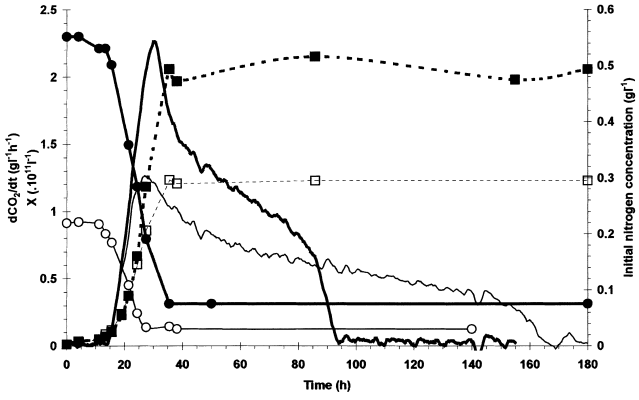


Figure 3. Changes in nitrogen concentration and cell population size during alcoholic fermentation in isothermal conditions (24°C). Two musts with different assimilable nitrogen concentrations (N_{init}). (1) N_{init} = 0.43 g/L: (▬) CO_2 production rate; (■) cell population; (●) residual nitrogen concentration. (2) N_{init} = 0.17 g/L: (—) CO_2 production rate; (□) cell population; (○) residual nitrogen concentration.

quantity of nitrogen assimilated by a single cell. We have assumed, for N_{ST}, that the activity of a single yeast may be determined from the amount of nitrogen assimilated alone. This part of the model was validated in two steps, the first involving fermentations *without* the addition of nitrogen, and the second involving fermentations *with* the addition of nitrogen.

We began by considering five experiments carried out with different amounts of initial assimilable nitrogen, in the range 0.07 to 0.57 g/L. If our assumption concerning sugar transporters was correct, then we could expect to observe homothetic (stacking when multiplied by a factor) activity curves as soon the rate of nitrogen assimilation by a single yeast cell was constant. This situation corresponds to the so-called stationary phase, in which the assimilable nitrogen of the must has been entirely depleted and the population remains constant. The activity curves were homothetic in this phase (Fig. 4), demonstrating that our model is consistent with the results of these experiments performed during the stationary phase. We calculated the homothetic factors for each piece of curve with respect to that corresponding to an experiment initiated with N_{init} = 0.17 g/L. This experiment was considered to be the reference fermentation, with a factor equal to 1. We used these homothetic factors to validate the model of the number of sugar transporters during the growth phase. If the model was correct, then dividing the activity curves of dCO_2/dt by the population $X(t)$ [see Eq. (3) with the corresponding parameters identified later in this study] and by N_{ST} (see the parameters given later), calculated from on-line measurements of assimilable nitrogen, would generate identical curves. These curves may be considered to estimate the mean activity v_{ST} of a single sugar transporter. We performed two specific experiments, with N_{init} = 0.17 g/L and N_{init}= 0.43 g/L, with both population size and assimilated nitrogen measured during the growth phase. The two estimated curves of dCO_2/dt divided by $X(t)$ and N_{ST} superimpose well (see Fig. 12). Thus, our

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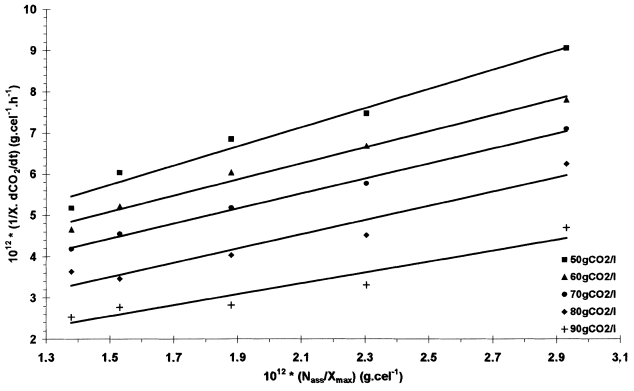


Figure 4. Activity values at various points in the stationary phase, for several musts fermented in similar conditions (24°C).

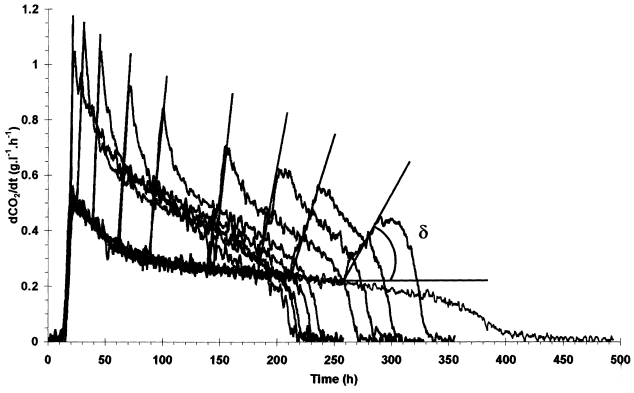


Figure 6. Changes in the rate of fermentation (δ) at various stages of fermentation: N_{init} = 0.07 g/L, addition of 0.3 g/L of $(NH_4)_2HPO_4$ at CO_2 = 0, 5, 10, 20, 30, 40, 50, 60, 70 g/L.

mathematical model was qualitatively able to restore the effect of assimilable nitrogen during the growth and stationary phases.

We then had to take into account the effect of adding nitrogen during the stationary phase. We compared the effects of adding equal amounts of nitrogen, in different ways (as a single or multiple additions, with various slopes for the rate of addition) and at different times during fermentation. In all cases, similar levels of activity were observed, after a transition phase of variable duration (Fig. 5). Indeed, the duration of the transition period was determined by the rate of nitrogen assimilation (see below). It should be noted that the amount of nitrogen devoted to the synthesis of transporters is greater during the stationary phase than during the growth phase (see later).

Dynamics of Nitrogen Assimilation: Nitrogen Transport

The extent to which added nitrogen increased reaction kinetics [slope (δ)] depended on the stage of the fermentation at which the addition was made (Fig. 6). The dynamics of this change seem to depend on the time required for the assimilation of nitrogen and, indeed, one of our assumptions

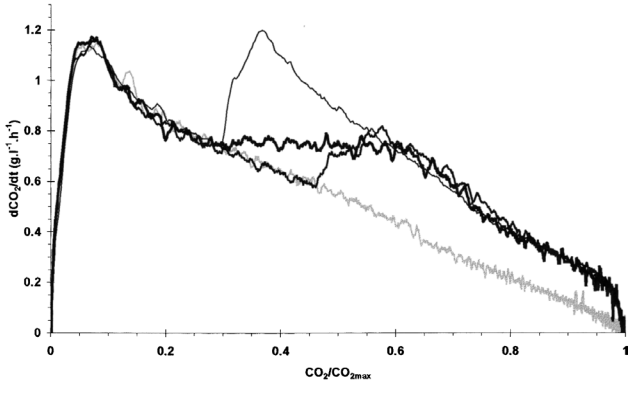


Figure 5. Effect of nitrogen additions [0.3 g/L of $(NH_4)_2HPO_4$] on the rate of CO_2 production. N_{init} = 0.17 g/L; S_0 = 280 g/L; (—) reference, (—) single addition at CO_2 = 40 g/L, (—) single addition at CO_2 = 60 g/L, (—) gradual addition at CO_2 = 40 g/L, to maintain a constant rate of reaction.

was that the time required for nitrogen assimilation determines the dynamics of yeast activity. The rate of assimilation clearly depended on the rate of increase in the number of transporters. This suggests that, during this phase, nitrogen transport is the principal step limiting the increase in yeast activity due to nitrogen (a situation similar to the limitation of yeast activity by sugar transport). Ethanol strongly inhibits nitrogen transport (Bisson, 1991; Leao and van Uden, 1983). More than one mechanism for this inhibition may be envisaged: depolarization of the membrane and changes in the configuration of the transporters and/or the proton pump (Bisson, 1991; Calderbank et al., 1985). These mechanisms may involve a direct interaction between ethanol and the proteins or changes in the lipid environment of the proteins, or both. Leao and van Uden (1983) reported that inhibition by ethanol of ammonium permease activity was noncompetitive, consistent with what has been found for glucose transport systems.

We checked that the rate of nitrogen assimilation always determined the rate of increase in yeast activity by running two experiments to determine the rate of nitrogen absorption (Fig. 7). In all cases, reactivation was found to be tightly linked to the rate of nitrogen absorption. Based on these results, we used the same type of equation as for glucose transport inhibition.

Although glucose transport is inhibited at lower concentrations of ethanol than ammonium permeases, at higher ethanol concentrations, the inhibition of ammonium uptake rapidly exceeds that of glucose (Leao and van Uden, 1983).

Importance of Temperature

Temperature affects the rate of fermentation, with higher temperatures giving faster fermentation rates and shorter fermentations. The membrane permeases are highly temperature-dependent, because changes in temperature provoke conformational changes in these molecules (Entian and Barnett, 1992). Thus, several studies of the effect of ethanol and temperature on yeast growth and fermentation have shown that, for temperatures of 15° to 35°C, glucose

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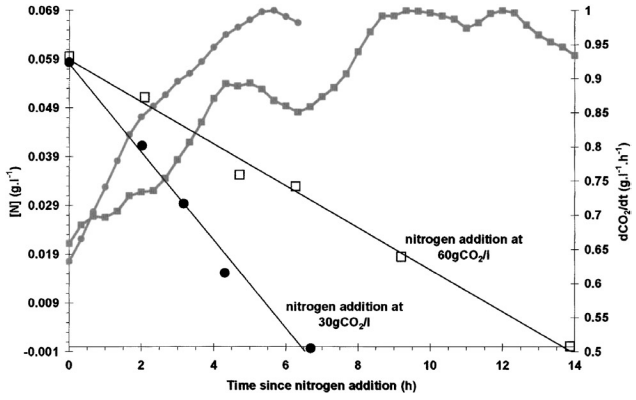


Figure 7. Nitrogen assimilation correlated with the increase in the rate of fermentation. The exhaustion of nitrogen corresponds to the maximum rate of reaction after the addition of nitrogen. The rate of nitrogen assimilation is a function of fermentation stage: N_{init} = 0.17 g/L; 0.3 g/L of $(NH_4)_2HPO_4$ added at CO_2 = 30 and 60 g/L.

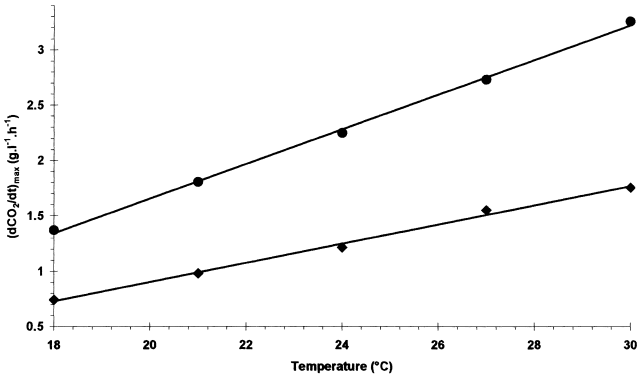


Figure 8. Effect of temperature on the maximal rate of CO_2 release in isothermal conditions: N_{init} = 0.17 g/L (♦) and N_{init} = 0.43 g/L (●); isothermal temperatures 18°, 21°, 24°, 27°, and 30°C.

transport and glycolytic flux increase steadily with temperature (Leao and van Uden, 1985; Sa-Correia and van Uden, 1983). Temperature has a similar effect on fermentation, confirming that glucose transport is the rate-limiting step in yeast activity. However, Arrhenius plots of the rates of xylose transport for various ethanol concentrations showed that inhibition of the glucose transport system is independent of temperature over the range 21° to 35°C (Leao and van Uden, 1982a).

Nitrogen assimilation has been shown to depend on fermentation temperature. Thus, even if the overall pattern of amino acid consumption remains constant over the range of temperatures used in wine production, low temperatures result in a lower rate of amino acid assimilation (Lopez et al., 1996), consistent with lower rates of fermentation and yeast growth.

Our experiments, with a single must fermented at different isothermal temperatures, are consistent with these observations. As reported by Bely et al. (1990a, 1990b), the rate of fermentation increases almost linearly with temperature (Fig. 8) and homothetically, giving curves with very similar general shapes and normalized superimposed curves (V/V_{max}).

Several kinetic models have already been described for isothermal conditions (Boulton, 1980; Bovee et al., 1984; El Haloui et al., 1989), but most enological fermentations are performed under anisothermal conditions.

The kinetics of such reactions may be very different from those of isothermal fermentations (Sablayrolles and Barre, 1989). We investigated whether this was the case by carrying out anisothermal experiments with two musts with different initial nitrogen concentrations (Fig. 9). We found that, when the temperature reached the value of the isothermal curve, the reactivation became more marked for the must with the largest amount of assimilable nitrogen, as previously reported (Sablayrolles and Barre, 1993). These results clearly demonstrate the existence of a ''memory effect'' or a residual effect of the initial temperature on the future activity of the yeasts. This suggests that a higher proportion of nitrogen is metabolized to generate glucose transporters if the temperature is low during the first part of the fermentation (i.e., when nitrogen is assimilated). We then extended our model to these conditions.

IDENTIFICATION AND PARAMETER ESTIMATION

The identification of the functions involved in the model [Eq. (1)] and the estimation of their parameters are difficult mathematical exercises due to nonlinearity, the strong interdependence of the variables, and the lack of sensors for on-line measurement of these variables. Parameters may be estimated in two different ways. The most classical method involves the use of specific methods to identify nonlinear dynamic systems. Many techniques are available for minimizing the quadratic error of estimation: the gradient method; the Gauss–Newton method; and the recursive least-squares method (e.g., Ljung and Soderstrom, 1983). Beyond the essential question of the identifiability of the parameters—which is not discussed here—the efficiency of such

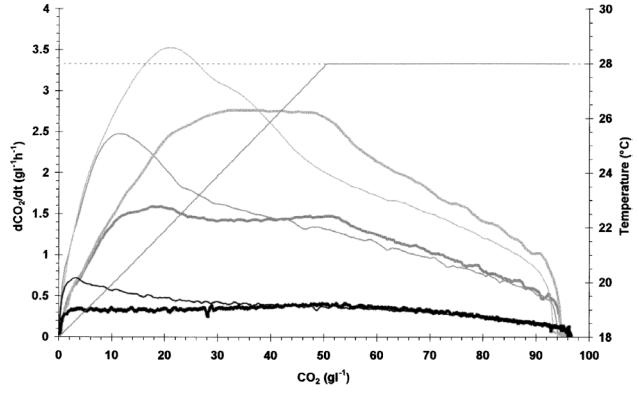


Figure 9. Effect of temperature on the shape of the kinetics curve in anisothermal conditions: (—) N_{init} = 0.07 g/L; (—) N_{init} = 0.28 mg/L; (—) N_{init} = 0.57 g/L; (—) isothermal conditions (28°C) and (—) anisothermal conditions (initial temperature 18°C, final temperature 28°C, rate of temperature increase 0.2°C (g CO_2/L)$^{-1}$.

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methods may be low if the initial values of parameter estimators are not close enough to the ''true'' unknown values. We therefore chose not to follow this classical procedure directly.

A second way of identifying parameters of model (1) depends on the specific structure of the model and the available measures. The aim is to break down the model to facilitate the separate estimation of the parameters of particular nonlinear functions. This second approach is well adapted to our problem as the nonlinear parts of the model are separable. We therefore carried out specific experiments to estimate simultaneously a limited number of parameters, by statistical computational methods and data analysis. Limiting the number of parameters to be estimated minimizes the difficulties of converging algorithms to global minima. This approach generates a set of estimated parameters. If this parameter set is unsatisfactory, it may be treated as a set of possible initial values for the more classical minimization method described earlier.

CO_2 released is measured and dCO_2/dt is calculated online. We have a few point measures of $X(t)$, and of $N(t)$ for some experiments. Mathematically, the evolution of the growth phase is exponentially unstable, making the estimation of parameters for this phase critical.

Identification of Yeast Growth Model

Two functions remain to be identified: (i) the maximum size of the population, $X_{\max}$, as a function of the initial assimilable nitrogen concentration; and (ii) the growth rate of the population in the exponential growth phase, k_1, as a function of temperature. The latency period, $t_{latency}$, was adjusted for each set of data. Its duration depends on the conditions at the start of fermentation.

We carried out certain experiments at $T = 24°C$, with initial assimilable nitrogen concentrations of 0.071 to 0.571 g/L, and other experiments at constant temperatures in the range 18° to 30°C, with constant N_{init}.

As stated earlier, the logistic model [Eq. (3)] is not explanatory for population growth. We therefore did not seek an explanatory relationship between k_1 and T, nor between $X_{\max}$ and N_{init}. These variables were found to be related by a linear function and a polynomial function, respectively: $k_1(T) = 0.0287\ T - 0.3762$ and $X_{\max}(N_{init}) = 10^9\ (-649\ N_{init}^2 + 698\ N_{init} + 7)$.

Figure 10 shows the good fit of the estimated population in two different experiments.

Identification of Number of Transporters per Cell (N_{ST}) in Isothermal Conditions

In the identification of the first equation of model [Eq. (1)], it is impossible to discriminate the number of hexose transporters, N_{ST}, from the numerator, $k_2(T)$, of the hexose transport function, v_{ST}. Without losing generality, we took $N_{ST} \times k_2$ to be 1 in the stationary phase of an experiment carried out at constant temperature, $T = 24°C$, with $N_{init} = 0.17$ g/L.

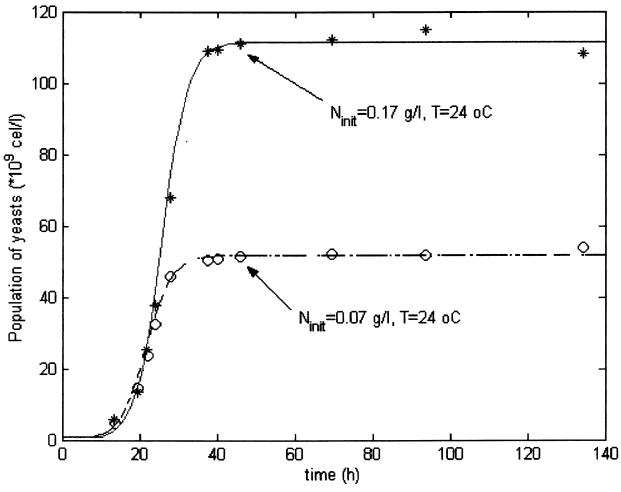


Figure 10. Estimation of yeast population size by a logistic model for two different initial nitrogen concentrations (N_{init} = 0.07 and 0.17 g/L, T = 24°C).

We ran four sets of experiments without nitrogen addition: two at constant temperature (24° and 28°C, respectively), with initial nitrogen concentrations, N_{init}, in the 0.071- to 0.571-g/L range. The other two sets of experiments were performed with initial nitrogen concentrations of 0.17 and 0.43 g/L, respectively, at a constant temperature in the range 18° to 30°C. For each set of experiments, the fermentation curves (i.e., $(\frac{dCO_2}{dt})$ plotted against CO_2) were superimposable in the stationary phase if they were multiplied by a translation factor. We then smoothed all the curves by nonparametric statistics to estimate the coefficients of translation of each part of the curve with respect to the corresponding curve for the experiment carried out at T = 24°C, with $N_{init} = 0.17$ g/L. These coefficients estimate the function, $N_{ST}(N_{\max}, X_{\max}(N_{init})T) \times k_2(T)$, for various values of constant T and N_{init}.

Based on these findings, we assumed that a bilinear function would best model $N_{ST} \times k_2$ in isothermal conditions (Fig. 11). Experimental points are denoted by asterisks in Figure 11.

The model we obtained for $N_{ST} \times k_2$ by the least-squares estimation method for nonlinear regression models (NLS2 library; see Huet et al., 1996) is as follows:

$$N_{ST}(N_{\max}(t) - N(t), X(t), T)k_2(T) = \lambda_a \frac{N_i(t)}{X(t)} + \lambda_b T(t) + \lambda_c \frac{N_i(t)}{X(t)} T(t) + \lambda_d \qquad (6)$$

where $N_i(t) = N_{init} + \lambda_0(N_{add})[N_{add}(t) - N(t)]$

with the following estimators: $\lambda_a = 335 \times 10^9$, $\lambda_b = 0.061$, $\lambda_c = 3 \times 10^9$, and $\lambda_d = -1$.

The function λ_0 is required only if nitrogen is added. In the absence of nitrogen addition and before such additions, $\lambda_0(0) = 1$. We found that $\lambda_0(N_{add}) = 3.2$ in experiments involving additions of 0.063 g/L nitrogen, late enough in the fermentation to prevent regrowth of the yeast population.

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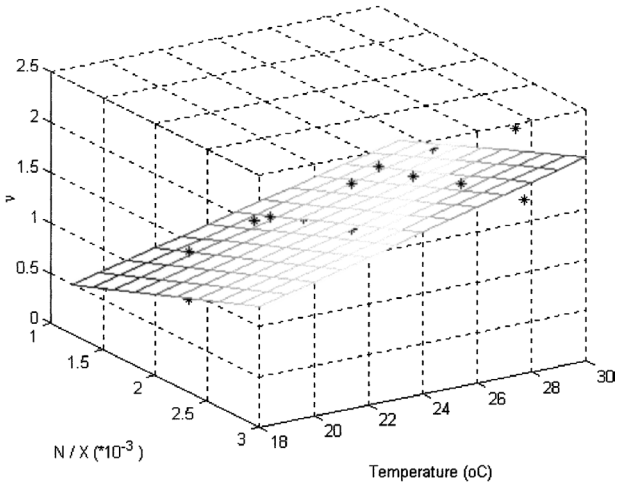


Figure 11. The product $N_{ST} \times k_2$ as a bilinear function of temperature and the nitrogen assimilated by a single yeast cell, in isothermal conditions, without nitrogen addition.

Before the addition of nitrogen, $N_i(t)$ may be considered to be the amount of nitrogen from the initial medium assimilated by a single yeast cell. If nitrogen is added to the medium, $N_i(t)$ is proportional to the added nitrogen assimilated by a single yeast cell. Because $\lambda_0(N_{add}) > 1$, it is clear that the yeast synthesizes more hexose transporters for a given amount of nitrogen if that nitrogen is added during fermentation rather than if it is present in the initial medium.

The quantitative validation of Eq. (6) in the growth phase is described in the section dealing with identification of v_{ST}. The identification of N_{ST} and k_2 in anisothermal conditions will be dealt in what follows. This model of $N_{ST} \times k_2$ is of the black box type. We are currently trying to develop an explanatory model based on genetic notions.

Estimation of the Model of Hexose Transport, v_{ST}

Five parameters must be estimated for the function v_{ST}. We examine this problem with experiments performed at a constant temperature of 24°C, thus the numerator of Eq. (4) may be considered as a single unknown parameter. The identification of v_{ST} is based on an inversion of the first equation of system Eq. (1) as follows:

$$\overline{v}_{ST}(S(t),E(t)) = \frac{v_{ST}(S(t),E(t),T)}{k_2(T)} \text{ at constant T} = \frac{d\mathrm{CO}_2(t)/dt}{X(t)N_{ST}(N_{\max}(t)-N(t),X(t),T)k_2(T)} \quad (7)$$

$$= \frac{\overline{k}_2\, S}{S + K_S + K_{SI}\, S\, E^{\alpha_S}} \quad (8)$$

The coefficient $\overline{k}_2$ is a ''normalizing'' parameter, based on a value for $N_{ST} \times k_2$ of 1 in the stationary phase of an experiment carried out at constant temperature, $T = 24$°C, with $N_{init} = 0.17$ g/L. Estimated values of $\overline{v}_{ST}$ are then calculated from the left side of Eq. (7) and from the estimation of $N_{ST} \times k_2$ calculated with λ_a, λ_b, λ_c, and λ_d, as shown earlier, and X as in Eq. (3) for specific experiments with nitrogen measurements. Two experiments with such measurements were carried out: one with $N_{init} = 0.17$ g/L and the other with $N_{init} = 0.43$ g/L. The curves of $\overline{v_{ST}}$ estimated from the data and Eqs. (3), (6), and (7) are plotted in Figure 12. These curves merge in both the stationary and growth phases, validating the assumption used and the identification of this part of the model.

The right side of Eq. (8) is a nonlinear regression model with unknown parameters that must be estimated from the curves of $\overline{v_{ST}}$. We used an ordinary least-squares estimation, based on the Gauss–Newton algorithm, and adapted to minimization under constraints. To overcome problems of identifiability, it was necessary to consider at least one experiment with ''normal'' initial conditions of glucose concentration, S_{init}, of about 200 g/L and another with ''extreme'' initial conditions of S_{init} of about 280 g/L (resulting in high ethanol concentration). The following estimators were obtained: $\overline{k}_2 = 1.5 \times 10^{-11}$, $K_S = 15$, $K_{SI} = 0.012$, and $\alpha_S = 1.25$. We actually estimated $\overline{k}_2 \times 10^{11}$, to overcome numerical problems arising from the calculation. We used different sets of initial values to start the optimization. No significant difference was observed. Figure 12 shows $\overline{v_{ST}}$ and the estimation of its parameters.

Identification of Number of Transporters, N_{ST}, and Function, $k_2(T)$, in Anisothermal Conditions

The rates of fermentation confirmed the law of Arrhenius in the range of temperatures used in enological conditions [see Fig. 8 and Ough and Amerine (1961)], yielding the following model for the function $k_2(T)$:

$$k_2(T) = \exp\left(-\frac{K_2}{T + 273.15}\right) \quad (9)$$

where the constant 273.15 converts the temperature, T, from degrees Celsius to Kelvin. By means of two experiments,

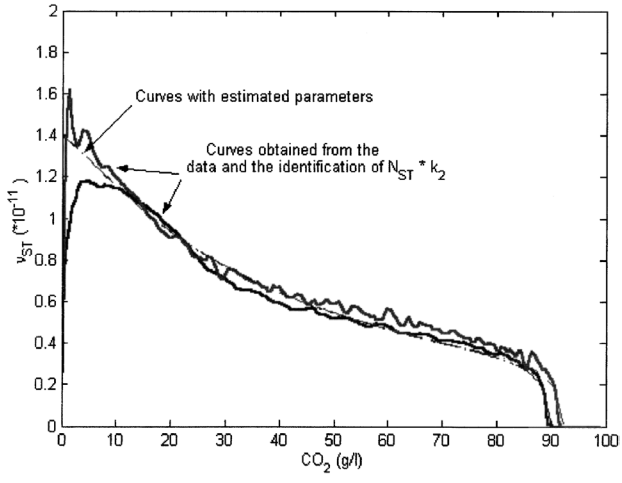


Figure 12. Identification of $\overline{v_{ST}}$ (S,E).

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one at constant temperature, T = 28°C, and the other in anisothermal conditions, with T varying from 18° to 28°C as CO_2 concentration increased from 0 to 50 g/L and with N_{init} = 0.17 g/L in both cases, we obtained K_2 = 7000.

For values of T in the 18° to 30°C range, the exponential curve of $k_2(T)$ in Eq. (9) is well approximated by a linear function of T. Given this fact, and in line with Eq. (6) of $N_{ST} \times k_2$ obtained in isothermal conditions, we decided to approximate the product of functions $N_{ST} \times k_2$ more generally by the following model:

$$N_{ST}[N_{\max}(t) - N(t), X(t), T_{ucd}(t)]k_2(T) = \lambda_1 T_{ucd}(t)N_i(t) + \lambda_2 T N_i(t) + \lambda_3 T + \lambda_4 T_{ucd} + \lambda_5 N_i(t) + \lambda_6 \quad (10)$$

where $N_i(t)$ has been introduced into Eq. (6) and $\lambda_1 = -0.36 \times 10^9$, $\lambda_2 = 3.3 \times 10^9$, $\lambda_3 = 0.09$, $\lambda_4 = -0.03$, $\lambda_5 = \lambda_a = 335 \times 10^9$ and $\lambda_6 = \lambda_a = -1$. Note that λ_i was estimated using Eq. (6), the estimated $\lambda_a \ldots \lambda_d$ and from the estimation of $\overline{v_{ST}}$ obtained previously in two anisothermal experiments, with N_{init} = 0.17 g/L and N_{init} = 0.43 g/L, respectively. Eq. (10) was validated in a third anisothermal experiment and described in what follows.

Identification of the Model of Nitrogen Assimilation, v_N

Function v_N describes the active transport of nitrogen by a cell—see second equation of Eq. (1). It is not possible to measure the concentration of residual assimilable nitrogen, N, directly on-line. The few point measurements we have are insufficient for direct determination of the dynamics of nitrogen consumption. We used an extended Kalman filtering method to estimate N and the parameters of v_N. This method is based on the measurement of CO_2 release and on all the other parameters of Eq. (1) previously identified (see earlier).

Thus, all the errors of modeling and estimation are transferred to this ultimate filtering step, resulting in a low overall quality of estimation. It was not possible to identify the function $k_3(T)$. We would have required much more precise measures of N to solve this problem. We have an approximation of this function only for the point T = 24°C. We did obtain parameters for: $k_3 = 10^{-12}$, K_N = 0.03, K_{NI} = 0.035, and α_N = 1.5. These values are satisfactory given the complexity of the model and the results obtained when simulating with all estimated parameters.

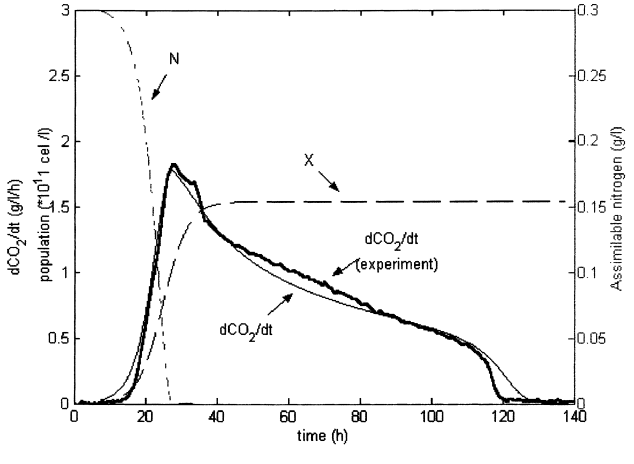


Figure 13. Simulation of a fermentation with S_{init} = 202 g/L, N_{init} = 0.28 g/L, and a constant temperature T = 24°C. Comparison with the corresponding experiment (—).

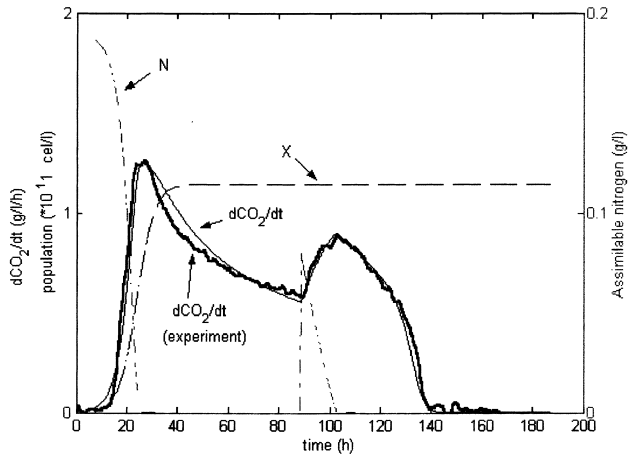


Figure 14. Simulation of a fermentation with S_{init} = 200 g/L, N_{init} = 0.17g/L, a constant temperature = 24°C and with the addition of 0.063 g/L of nitrogen at CO_2 = 60 g/L. Comparison with the corresponding experiment (thick line).

QUANTITATIVE VALIDATION

The purpose of this section is to validate the model identified earlier by testing its ability to predict experiments that were not used in the construction and estimation phases. Toward this aim, we compared the results of three specific new experiments, with the results of Eq. (1) simulated using

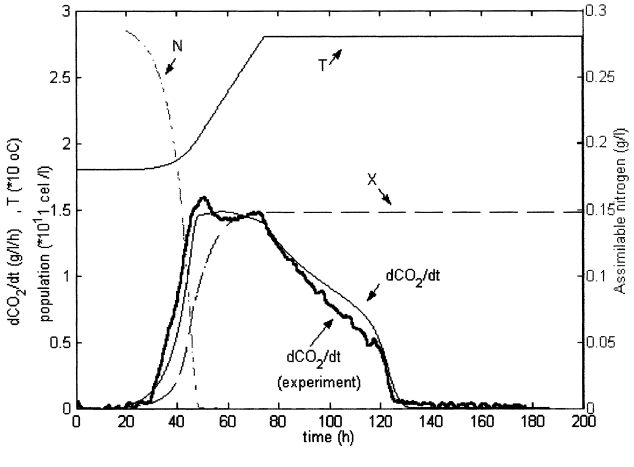


Figure 15. Simulation of a fermentation with S_{init} = 208 g/L, N_{init} = 0.28 g/L, and a temperature T varying from 18° to 28°C as the amount of CO_2 released increases from 0 to 50 g/L. Comparison with the corresponding experiment (thick line).

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MATLAB software in the same specific conditions. The conditions of these three comparative tests are as follows: The first experiment corresponded to isothermal conditions, at $T = 24°C$, with no added nitrogen, and the results are presented in Figure 13. Changes in calculated nitrogen content, N, population, X, and rate of CO_2 production, dCO_2/dt, are presented, together with the curve for CO_2 production from the corresponding experiment. The simulation fits the results for the real experiment very well.

Figure 14 shows the experimental curves of a fermentation performed in isothermal conditions, with the addition of 0.063 g/L nitrogen when CO_2 release reached 60 g/L. The simulated curve for dCO_2/dt fits very well the experimental curve.

The third comparison test concerns a fermentation in anisothermal conditions. The curve for temperature, T, is also given in Figure 15. The two curves (real and simulated) for dCO_2/dt fit less well in the growth phase than did the curves for the other simulations. However, this may be considered a good result given the complexity of the exponential phase with both T and N varying over time.

CONCLUSIONS

Our model for yeast alcoholic fermentation is the first taking into account the two main parameters limiting fermentation kinetics in wine-making conditions, specifically temperature (both in isothermal and anisothermal conditions) and assimilable nitrogen (including additions at different moments of fermentation). This model is based on physiological mechanisms. Its effectiveness with respect to experimental data, tested in very different conditions (Table I), points to the utility of this model for: (i) validating the main physiological mechanisms limiting fermentation kinetics, especially sugar transport, nitrogen transport, and inhibition by ethanol; and (ii) fermentation control—it is being integrated into a new control system for wine making, taking into account heat exchanges, to optimize the use of tanks and the energy required to cool the entire winery.

Table I. Comparison of fermentation duration when CO_2 released reaches $[(CO_2)_{max} - 2$ g/L] for experiments and their simulated curves.

Experiment	N_{init} (10^3 g/L)	Abs. (relative error) (%)
1	171	2.6
2	71	4.8
3	299	2.1
4	571	5
5	171	2.2
6	428	0.8
7	100	4
8	71	8.7
9	171	1.5
10	285	2.2
11	428	3.9
12	571	4
13	171	4.9
14	285	3.1
15	428	4.6
16	571	6.6
17	171	4.3
Mean error (%)		3.8

NOMENCLATURE

α_N	constant
v_N	function describing the active transport of nitrogen per a cell (g/h)
α_S	constant
v_{ST}	function describing the active transport of sugar per a sugar transporter (g/h) before $X_{max}/2$, $T_{ucd}(t) = T(t)$, after $X_{max}/2$, T_{ucd} = constant
$CO_2(t)$	carcon dioxide released (g/L)
$E(t)$	ethanol concentration (g/L)
k_1, k_2, k_3	functions depending on temperature
K_N	Affinity constant of yeast cells for nitrogen transport (g/L)
K_{NI}	Constant for the inhibition by ethanol of nitrogen transport in yeast cells (L/g)
K_S	affinity constant of sugar transporters (g/L)
K_{SI}	constant for the inhibition of sugar transporters by ethanol (L/g)
$N(t)$	residual assimilable nitrogen concentration (g/L)
$N_{add}(t)$	concentration of added assimilable nitrogen if any addition was made (g/L)
N_{init}	initial assimilable nitrogen concentration (g/L)
$N_{max}(t)$	$N_{init} + N_{add}(t)$ (g/L)
N_{ST}	number of sugar transporters in a cell
$S(t)$	residual glucose concentration (g/L)
S_{init}	initial glucose concentration (g/L)
$T(t)$	fermenting must temperature (°C)
$t_{latency}$	time at which cell division begins (h)
$T_{ucd}(t)$	temperature until the last cell division; that is, when $X(t) = X_{max}/2$ (°C)
$X(t)$	cell population size (/L)
X_{init}	initial cell population size (/L)
$X_{max}(N_{init})$	maximum population size during the stationary phase (/L)

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